



DEPARTMENT OF MECHANICAL

ENGINEERING

HACKSIMUL8

A 8 HOUR INTERDISCIPLINARY HACKATHON ON
SIMULATION USING MATLAB AND SIMULINK

Problem Statement

Research and development activities related to UAV quadcopters have observed a considerable surge from past few years. Today, quadcopters are widely used in military and law enforcement, disaster monitoring, package delivery, monitoring, search and rescue operations etc. Quadcopters are also used by researchers in various fields of control theory and robotics for implementation and evaluation of new ideas. Quadcopter is highly non-linear system because of coupled dynamics and uncertainties due to environmental disturbances. Therefore, a robust control system is required for stable flight of quadcopter.

Proportional-Derivative-Integral (PID) controller is mostly used for control of physical systems due to its simplicity and easy implementation. The accuracy of PID controller depends upon the gains, better the gains, and better are the results. However, highly non-linear systems subjected to system parameters' variations and environmental disturbances mitigate the performance of PID controller. For such nonlinear systems conventional off-line PID tuning techniques are not very effective. Online tuning of PID parameter is required for better performance of such nonlinear systems.

Task 1 – Develop the dynamic mathematical model of the 6 DOF UAV Quadcopter in MATLAB and SIMULINK in terms of state space equations.

The structural diagram of the UAV quadcopter and the coordinate systems that can be used in building the dynamic mathematical model of the UAV quadcopter are shown in Fig. 1. In Fig. 1, the UAV's angular velocity, torque, and the corresponding force generated by the four propellers, driven by four electrical motors from 1 to 4 (also called rotors). Assume that

- i) The quadcopter model is a homogeneous and symmetric block.
- ii) Mathematical model built on a cross-shaped quadcopter, the center of the quadcopter is the center of gravity;
- iii) Neglecting elasticity, consider the transmission system absolutely rigid.

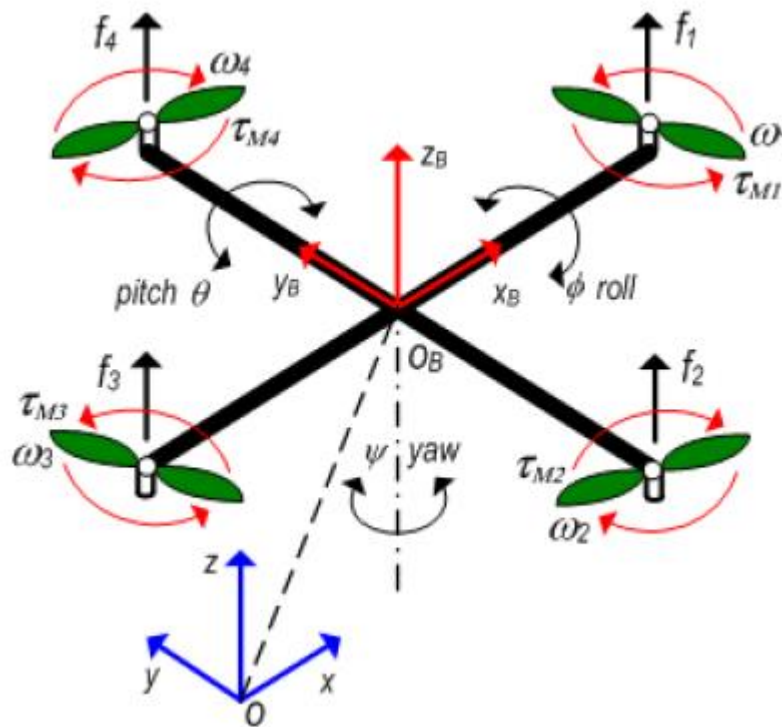


Fig. 1 The UAV quadcopter structure in the body frame and ground coordinates

The UAV's absolute translational position in the inertial frame x, y, z -axes, attached to the ground, is denoted $\xi, \xi = [x \ y \ z]$. The angular position is determined by the Euler angles, $\eta, \eta = [\phi \ \theta \ \psi]$. The roll angle ϕ determines the rotation of the UAV quadcopter around the x -axis. The pitch angle θ determines the rotation of the UAV quadcopter around the y -axis. The yaw angle ψ determines the rotation of the UAV quadcopter the z -axis.

Task 2 – Design the Altitude Control System for 6-DOF UAV Quadcopter using PID Control

Requirement – Neat description of the methodology used for tuning of PID controller gains should be presented during the evaluation. Use the best method available for tuning.

Task 3 – Generate the data from the simulation experiments of the altitude control system and train a suitable ML model for self-tuning of PID gains

Task 4 – Implement the ML model and present test cases with suitable comparisons and merits of the utilised ML model

The parameters of the 6-DOF UAV quadcopter are presented as shown in Table 1.

Table 1. Parameters of the UAV quadcopter

Parameter	Symbol	Value
Quad. mass	m	0.516 kg
Arm length	l	0.225 m
Gravity	g	9.81m/s ²
Inertia moment of the rotor	I_M	3.368e-5kg.m ²
Thrust factor of rotor	k	2.996e-6N.s ²
Drag coeffi.	b	1.260e-7 N.m.s ²
Inertial constants	I_{xx}, I_{yy}	4.984e-3kg.m ²
	I_{zz}	8.958e-3kg.m ²

Reference –

Mien, T., & Tu, T. (2024), “Design and Quality Evaluation of the Position and Attitude Control System for 6-DOF UAV Quadcopter Using Heuristic PID Tuning Methods”, International Journal of Robotics and Control Systems, 4(4), 1712-1730, doi:<https://doi.org/10.31763/ijrcs.v4i4.1594>